Shivesh Singh

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SUMMARY

Senior Data Engineer (GenAI) building agentic AI systems and data-ops automation in production. Designed multi-agent orchestration frameworks and shipped LLM workflows integrated with enterprise tooling (ServiceNow/Confluence), Databricks, lineage systems, and pgvector-backed retrieval. Led small teams and owned delivery across COE (new product engineering) and COC (client solution delivery and optimization).

Education

Indian Institute of Technology (IIT), Mandi — *2023 – 2025*

M.Tech (Research) in **Machine Learning**

Thesis: Machine Learning Approaches for Spatiotemporal Modelling of Environmental and Demographic Changes

REC Kannauj — *2019 – 2023*

B.Tech in **Computer Science & Engineering**

Experience

LTIMindtree — *July 2025 – Present*

Senior Data Engineer (GenAI / Agentic AI Systems) — *Noida, India*

- **Architected a DeepAgents-based orchestration layer** for planning, task delegation, specialized routing, memory and policy constraints, and tool abstraction across enterprise AI and DataOps workflows.
- **Built and refined support-persona agent workflows** using Azure OpenAI, LangGraph, semantic retrieval, and trace-driven observability to improve reliability, debuggability, and operational readiness.
- **Expanded Thrive.ai into a ServiceNow-integrated AI Ops framework** with incident updates, similar-incident retrieval, RCA recommendations, closure-note generation, approval loops, and dashboards for SLA, aging, volume, and recurrence.
- **Enabled secure Databricks integration for AI Ops scenarios** across Hive Metastore, Unity Catalog, OAuth-based access patterns, and validation workflows for enterprise data checks.
- **Designed GenAI automation with ServiceNow and Confluence** for intake, analysis, BRD drafting, runbook generation, and resolution recommendations with reliability checks and human review built in.
- **Guided implementation across delivery teams,** translating ambiguous requirements into technical designs, reusable workflow patterns, and production-ready components while mentoring engineers through implementation decisions.
- **Contributed to BlueVerse and DAAN GenAI CoE initiatives** by building sub-agents for the Data Consumption persona, equipping them with domain-specific tools from integrated MCP servers and web search, and pushing agent workflows to the BlueVerse marketplace for client use cases including ShakeShak and Paramount.

Selected Projects

Sound Scene Synthesis — DCASE Challenge 2024 — *shivesh235/sound_scene_synthesis*

- **Achieved 4th place globally** at DCASE Challenge 2024 Task 7 by fine-tuning AudioLDM on a custom 3-source audio corpus combining ESC50, AudioCaps, and the DCASE Acoustic Scene Dataset across 10+ scene categories.
- **Engineered a foreground/background audio mixing pipeline** with amplitude weighting (foreground $2\times$, background $0.5\times$) to synthesize contextually layered soundscapes that reflect real-world acoustic relationships.
- **Trained a separate VQ-VAE model for foley sound generation,** extending the generative pipeline beyond AudioLDM to cover diverse environmental and event-driven sound types.
- **Built a Flask web UI** for music synthesis conditioned on facial expression, user age, and dominant image colors extracted from uploaded photos, demonstrating multimodal prompt-driven audio generation.

Rerender A Video: Zero-Shot Text-Guided Video-to-Video Translation — *shivesh235/Rerender_v2v*

- **Built a zero-shot text-guided video translation framework** on ControlNet (HED and Canny edge maps) and Stable Diffusion, enabling arbitrary style transfer to video without any domain-specific retraining.

- **Enforced global and local temporal consistency** via GMFlow optical flow estimation with hierarchical cross-frame constraints on shape, texture, and color coherence throughout the diffusion sampling process.
- **Extended the pipeline with NinaSR super-resolution and MIRNet low-light enhancement** to improve output quality for degraded and night-time footage, and integrated AudioLDM for audio adaptation synced to translated visuals.
- **Exposed the full pipeline via a Gradio web UI** with configurable ControlNet strength, denoising, resolution, and enhancement options; supported LoRA-based style customization for user-defined outputs.

Phi-4 Medical Visual Question Answering Fine-tuning

shivesh235/phi-4-finetuning

- **Fine-tuned Microsoft Phi-4 Vision (32K-context multimodal LLM)** on the VQA-RAD medical imaging dataset using LoRA (rank 16, alpha 32) targeting all attention projection layers (q/k/v/o), enabling efficient adaptation with minimal trainable parameters.
- **Configured multi-GPU distributed training** with PyTorch Lightning DDP, BF16 mixed precision, 4-step gradient accumulation, and gradient clipping for stable convergence over 10 epochs with early stopping.
- **Built a comprehensive evaluation suite** covering exact-match accuracy, BLEU/ROUGE scores, and specialized binary accuracy for closed-ended clinical yes/no questions, with WandB integration for full experiment tracking.

PUBLICATIONS & AWARDS

- **Publication:** Shivesh Singh, Nitu Kumari, Aditya Nigam. “Deep spatio-temporal modeling of vegetation index and population density in Himachal Pradesh, India.” *Spatial Information Research*.
  DOI: 10.1007/s41324-025-00622-3
- **Awards:** 4th Rank, DCASE Challenge 2024; AI Innovation Hackathon, Yamaha Motor Solutions